VISHAL KUMAR

Senior Software Engineer and Tech Lead, Amazon Last Mile Maps

Bellevue, WA | vishal4chelsea [at] gmail [dot] com | linkedin.com/in/vishalnitt | github.com/vishal4c

SUMMARY

Engineer with 15+ years in large-scale distributed systems, the last nine in geospatial data at Amazon. I lead the architecture behind Amazon's maps: the pipelines that produce the visual and routing map artifacts behind millions of daily delivery routes, and the ML systems that learn from driver sensor data. I drive Amazon-wide adoption of the OpenStreetMap ecosystem and open-source geospatial frameworks and tooling. Current research applies LLM and vision-language agents to map editing, conflation, and geospatial data labeling, with deterministic verifiers on every agent edit. Author of peer-reviewed work at ACM SIGSPATIAL; speaker at State of the Map US and FOSS4G NA.

PUBLICATIONS

Scalable Conflation for Maps Data Replay between Heterogeneous Geospatial Data Sources. ACM SIGSPATIAL 2026, Industrial Track. Riverside, CA, Nov 2026.

MapScout: An Agentic Harness for Map Editing and Geospatial Data Labeling. ACM SIGSPATIAL 2026, Demonstration Track. Riverside, CA, Nov 2026.

Deep Classification of Frequently-Changing Activities from GPS Trajectories. ACM SIGSPATIAL Workshop on Computational Transportation Science (IWCTS 2022). Best Paper Award. doi:10.1145/3557991.3567784. Proceedings: dl.acm.org/action/showFmPdf?doi=10.1145/3557991. Companion open dataset: GOAL, GPS Ordered Activity Labels (github.com/amazon-science/goal-gps-ordered-activity-labels).

TALKS AND WORKSHOPS

- 2026 **Workshop:** Routes to Safety: A Wildfire Evacuation Map from Open Data. FOSS4G North America 2026, Sacramento, Nov 2. Repo: github.com/vishal4c/routes-to-safety-workshop
- 2026 **Talk:** Accelerated Multi-Source Vector-Tile Generation with Open-source Frameworks. FOSS4G North America 2026, Sacramento, Nov 4.
- 2026 **Talk:** How Amazon Generates Map Tiles for Last-Mile Delivery Using OSM-Derived Datasets and Planetiler. State of the Map US 2026, Madison, Jun 12.
- 2022 **Paper presentation:** Deep Classification of Frequently-Changing Activities from GPS Trajectories. ACM SIGSPATIAL IWCTS 2022, Seattle.

PUBLISHED ARTICLES

- 2026 Agentic AI's Next Frontier Is the Physical World. Trust Will Decide Who Wins. The AI Journal, 2026.
- 2026 The Agent Drafts, a Human Approves: Inside State of the Map US 2026. HackerNoon, 2026.
- 2026 How Open Source Runs the Mapping World. HackerNoon, 2026.

AWARDS

- 2025 Innovation Award, Amazon Last Mile Geospatial All Hands, for the GeoAI multi-agent framework
- 2025 People's Choice Award, Amazon GenAI Hackathon (GraphFix AI Agent: validates map data by comparing routes across map providers)
- 2024 Special Award for Operational Excellence, Amazon Geospatial Hub Winter Hackathon (Oncall Whisperer)
- 2023 1st Place, Amazon GS Hub Hackathon (driver building entry and exit events from raw GNSS signals)
- 2022 Best Paper Award, ACM SIGSPATIAL IWCTS 2022 (Deep Classification of Frequently-Changing Activities from GPS Trajectories)
- 2021 OE and Quality Tech Award, Amazon Maps, Locations, and TTO Hack Day (Route Timeline Replay Tool)
- 2019 Best Overall Award, Amazon Last Mile HackDay (cell-based road network generation for faster maps artifact deployment and maps defect isolation)
- 2019 Innovation Maven Award, Amazon Last Mile, for the Spark migration of worldwide map ingestion

EXPERIENCE

Senior Software Engineer and Tech Lead, Amazon Last Mile Maps 2020 to present, Bellevue, WA

- **Agentic mapping (2025 to present).** Lead the GeoAI agent framework on AWS Bedrock AgentCore, used by 1,000+ internal partners and featured in Amazon's Sept 2025 announcement of AI-powered tools for delivery partners. Built MapScout, an agent harness that lets LLM and vision-language agents edit maps and label geospatial data from satellite imagery, with deterministic verifiers gating every edit. Applying the same pattern to map conflation across providers and to sidewalk and walk-path detection.
- **FlashTileGenEngine (2025 to 2026).** Architected a database-free vector tile pipeline (Planetiler, GeoParquet, Apache Sedona on Spark) that replaced the PostGIS stack. Worldwide tile refresh dropped from 31 hours to 4, about 70 percent faster, and roughly $480K per year of database cost was removed. 22 logistics-specific layers now serve delivery planning in 19 countries.
- **Open data adoption.** Led the migration of Amazon's delivery-maps systems onto Overture Maps data, schema, and GERS identifiers. Designed the Airflow pipelines that turn commercial datasets into open formats and specified the 11 MapMaking APIs behind partnerships with HERE Technologies and the Overture Maps Foundation.
- **Trajectory ML (2020 to 2024).** Productionized a bi-directional LSTM with attention that classifies driver activity from GPS traces alone. It replaced heuristic activity inference across worldwide delivery operations in 2024 and was published at ACM SIGSPATIAL IWCTS 2022 with an open dataset.

Software Engineer, Amazon Last Mile Maps 2016 to 2020, Seattle, WA

- Designed the Spark pipeline that generates Amazon's routable road network; migrated the monolithic 48-hour worldwide map ingestion to an 8-hour distributed system, with a 6x speedup on the continental critical path.
- Prototyped and led the cell-based partitioning design that cut global map update cycles from 2 to 4 days to under 15 hours; led external design reviews adopted across multiple teams.
- Led ingestion from multiple third-party map providers and the technical coordination for map launches in 10+ countries.

Software Engineer, Amazon 2015 to 2016, Seattle, WA

- Launched content generation pipelines for Japan and China; cut map data refresh from 30 days to 2 days.

Software Engineer, Amazon 2011 to 2015, Chennai

- Built the OCR-based content generation pipeline for international Kindle markets, including paragraph detection and automated footnote linking; patent application on improved character recognition for Japanese script.

Research Engineer, CDOT Alcatel-Lucent Research Center 2010 to 2011, Chennai

- WiMAX network management applications; kernel modules for IP tunneling; bandwidth management for carrier-grade networks.

Application Developer, Girnarsoft 2009, India

- J2EE distribution-management and insurance applications for enterprise clients.

COMMUNITY AND SERVICE

- Overture Maps Foundation: participant in the Schema and Buildings task forces.
- Senior individual contributor in a 34-engineer geospatial organization; member of its promotion review panel; 300+ technical interviews conducted.
- Mentor to engineers and researchers in big data and applied AI, inside Amazon and through ADPList.

TECHNICAL SKILLS

Big spatial data	Apache Spark, Apache Sedona, PySpark, Airflow, DuckDB, GeoParquet, PostGIS, Hadoop
Maps and routing	Planetiler, PMTiles, MapLibre GL, OSRM, OpenStreetMap, Overture Maps and GERS, map matching, OpenLR, linear referencing
AI and ML	AWS Bedrock AgentCore, Strands Agents, MCP, LLM and VLM agent harnesses, PyTorch, LSTM sequence models
Cloud	AWS (EMR, Lambda, Step Functions, Glue, SageMaker, ECS), CDK, event-driven and microservice architectures
Languages	Python, Java, Scala, SQL, C++

EDUCATION

B.Tech, Computer Science and Engineering, National Institute of Technology, Tiruchirappalli (NIT Trichy), India, 2009. GPA 8.60/10.